

Big Data and Automated Content Analysis

Final Individual Project (65%)

Academic Year 2026/27

1 Final individual project

The final individual project accounts for 65% of the overall course grade. Students design and execute an independent Computational Communication Science project that demonstrates their ability to formulate a research question, collect and analyse data, and apply fundamental techniques taught in the course.

This is an individual assessment. The report and all submitted code must be the student's own work, except where external sources or permitted generative-AI contributions are appropriately disclosed in accordance with the course manual.

Your submission must include:

1. a PDF report; and
2. a .zip folder containing all relevant code and output files.

PDF report

The recommended length of the report is five to six pages. It should include the following components.

A. **Introduction** (one to two pages)

- A clearly formulated research question;
- Any subquestions and/or hypotheses;
- References to relevant literature, including course readings where appropriate.

B. **Analytic strategy**

- A transparent overview of the methodological approach;
- A justification for the choice of techniques, referring explicitly to methods discussed during the course.

C. **Dataset description**

- A carefully collected and relevant dataset of non-trivial size, for example retrieved via APIs or constructed by combining existing datasets;
- An exploratory description of the dataset;
- A brief argument explaining why the dataset is suitable for the research question.

D. **Results**

- A clear presentation of the analyses and findings needed to answer the research question;
- Appropriate tables, figures, examples, or model-evaluation results;
- A substantive interpretation of the findings that distinguishes empirical results from speculation.

E. Conclusion and critical reflection

- A well-substantiated answer to the research question;
- A critical discussion of the strengths, limitations, and uncertainties of the research design, data, analysis, and conclusions;
- An explanation of the implications of these limitations for the interpretation of the findings, as well as feasible directions for future research.

The conclusion and critical-reflection component contributes to the assessment of ILO G: students critically discuss the strengths and weaknesses of their completed research and suggest improvements.

Project files (.zip folder)

Alongside the PDF report, include a compressed folder containing:

- Python scripts or notebooks for data collection, preprocessing, analysis, and visualisation. These files must be well documented, reproducible, and tailored to the student's own project;
- Output files, such as tables, plots, or summary statistics;
- The data needed to reproduce the analysis, where sharing those data is legally, ethically, and contractually permitted;
- A README file describing:
 - how to run the code;
 - what each script or notebook does; and
 - any software or library requirements.

Requirements and guidance

Depending on the chosen topic, students must apply at least one technique taught in the course. Projects should include an exploratory description of the dataset and apply fundamental analytical techniques.

During lab sessions and individual consultations, students may discuss:

- the scope of the project;
- the methodological requirements that the project must meet; and
- how the chosen analytical methods contribute to the final grade.

Important notes

- All external data sources and code, including libraries, templates, and tutorials, must be properly cited.
- Code must be tailored to the student's own project and run without errors.
- Any non-trivial contribution by a generative-AI tool must be disclosed in the relevant notebook or script, as specified in the course manual.

- The rules on generative-AI use in the course manual apply in full. In particular, students may not upload course datasets or full analysis notebooks to external generative-AI services, or use generative AI as a black-box analysis engine for graded work.
- Students are free to select any topic of interest—including topics outside traditional communication science or the social sciences—provided that it is framed around a research question and based on original data analysis.

Submission instructions

Submit both the PDF report and the `.zip` folder containing code and outputs via Canvas.

Deadline: Submission is open until 16 December, 23:59. Late submissions are handled in accordance with the course manual and applicable examination regulations.